UM Enablement: AI Enabled UCX

Project Description/Context:

InQbator AI aims to improve EviCore clinical review workflows with the addition of AI-enabled features designed to make the navigation of clinical documentation faster and easier. These features have been designed and tested with internal MD's and include the following: creating a unified PDF to reduce the number of clicks and amount of flipping between browser tabs; using AI to classify document types (e.g. clinical notes, imaging, labs, etc.), enabling clinicians to tap into their strong intuition about what kind of documentation is relevant; using AI to extract date information from clinical documents to help clinicians better understand the freshness of the information it contains; creating a table of contents to make it easier to navigate through long documents; using AI to highlight relevant information on PDFs, to draw clinical reviewers attention to key pieces of information; removing exact duplication files from the unified PDF – and various other features that have been requested by our clinical partners.

Solution Overview

The purpose of the project is to reduce the clinical review time for the MDs by increasing the speed at which they can retrieve the information from the clinical documents . Below is high level overview of the tasks we need to perform for the options considered

-
  - Develop a consumer for the Kafka event `XPLERSSCaseSaved` that monitors for `AuthStatus` values O, I, and R. This consumer will invoke OV to retrieve all documents linked to the case, excluding notifications. Filters can be applied based on the object class when fetching from OV.
  - Merge all OCR-processed documents into a single document. Utilize AI to determine the document type, highlight pertinent information, and upload the newly consolidated document back to OV with the updated object class.
  - Continuously listen for the OCR scan completion event. If any additional documents are received, retrieve the previously combined document from OV, create a new document, and perform a soft delete of the old document by issuing a delete command to OV.
  - Update the docgen to ensure that the case file and case information report documents do not fetch documents belonging to the newly created object class.

- Modify the docgen web form and file manager service to ensure that the newly created documents appear in the incoming fax section and have the "View OCR" button enabled.
- Adjust UCX to ensure that the new documents are displayed correctly.

Scope

In-Scope

- Displaying the combined document in UCX and CDP
- Storing the combined document
- Combine all the OCR documents attached to a PA request into one
- Highlight the relevant information in the document

Out of Scope

- Any clinical portals such as ISAAC , eP , PAC manager are out of scope for this
- There is an opportunity to improve the current implementation of display of OCR document, but it is out of scope for the estimation
- Inqbator work is out of scope for estimation

Assumption

Team Impacted

-
  - **XPG**: Onboard new service to fetch the documents from ObjectValet
  - **CDP**: Conduct testing to ensure that the combined documents are displayed correctly.
  - **UCX**: To call the clinical review Inteliigence service and integrate CRI front end.

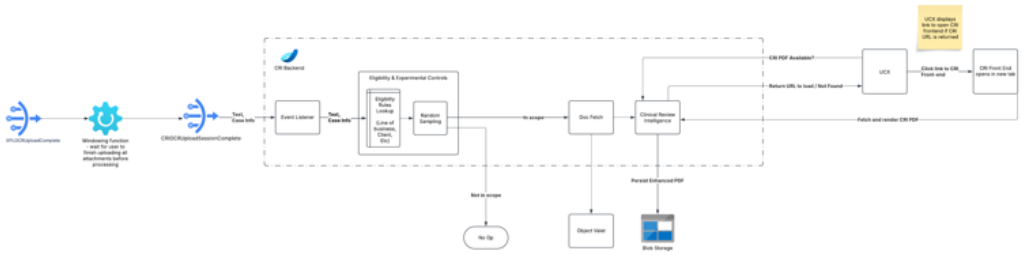
Risks

- On average, there are 3.5 documents attached to each case, with 80% of these documents being fewer than 10 pages long. Based on these metrics, the total combined document length is estimated to be around 40 pages. As these documents are downloaded for display purposes, we may experience some latency due to the increased size of the documents.
- The largest document we have encountered in the past year was 440 MB, and approximately 10% of documents exceed 200 MB in size. These larger documents pose a risk of increased

download times.

- Additionally, over 15% of documents are longer than 20 pages, which raises concerns about potential latency during the download process when these documents are combined.

Solution Diagram



T-Shirt Estimation

Labor Cost :

Blended cost : $150,000

Non Labor cost :

CRI blob storage cost : $12000/yr

CRI API infra cost : $5000/yr

On going cost : $25000/yr

Total Build cost for year 1 : $97000

Total infra cost for year 2 onwards : 25000/yr

T shirt size : X-Small

-Shirt Sizes	High		Average		Timespan Estimate
	Hours	Cost	Hours	Cost	
XX-Small	1250	$100,000	750	$60,000	1 Sprint
X-Small	2500	$200,000	1,875	$150,000	2-3 Sprints

Small	6250	$500,000	4,375	$350,000	1-1.5 Pis
Medium	12500	$1,000,000	9,375	$750,000	2-4 Pis
Large	37500	$3,000,000	25,000	$2,000,001	1yr + (5Pis)
X-Large	62500	$5,000,000	50,000	$4,000,000	1yr + (multiple teams)
XX-Large	125000	$10,000,000	93,750	$7,500,001	Multi-year